



Carisurg

Assignment 19

Final Documentation

Due Date: July 21st, 2026.

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Verdict

No. Neither Random Forest nor Gradient Boosting should replace the Week 6 logistic regression baseline: both trade away the one number that matters most in a triage system — recall on the most critical patients (ESI 1) — in exchange for a higher overall accuracy score, and that is not a trade worth making.

Setup & Data Decisions

This is the same 55,121-encounter Yale EMMLC triage extract used since Week 5, the same top-10 feature shortlist carried over from the Week 5 memo (age, SpO2, chest pain, shortness of breath, suicidal ideation, alcohol intoxication, altered mental status, ambulance arrival, respiratory rate, heart rate), and the identical 80/20 stratified train/test split (`random_state = 42`) used in Week 6. No features changed this week — any difference in the numbers below comes from the model, not the data.

Two complex candidates were built and evaluated against the Week 6 baseline:

1. A **Random Forest** (300 trees, `class_weight="balanced"`)
2. And a **Gradient Boosting** model (`HistGradientBoostingClassifier`, 300 boosting rounds).

A third variant — Random Forest **tuned specifically for macro recall** via `RandomizedSearchCV` — was added to test whether the weakest result below was a fixable hyperparameter problem or something more fundamental.

Benchmark Table

Model	Accuracy	Macro Recall	ESI-1 Recall	Weighted F1	Macro F1	Train (s)	Infer (ms/pred)	Interpretability
<i>Logistic Regression (baseline)</i>	0.254	0.417	0.75	0.327	0.214	0.337	0.0002	High
<i>Random Forest</i>	0.498	0.282	0	0.492	0.285	29.094	0.1218	Medium
<i>Random Forest (tuned for recall)</i>	0.397	0.35	0	0.423	0.268	146.826	0.0294	Medium
<i>Gradient Boosting</i>	0.341	0.405	0.375	0.389	0.253	0.657	0.0061	Low

Three Arguments For

1. **Random Forest nearly doubles overall accuracy** (0.498 vs 0.254) and improves macro F1 (0.285 vs 0.214) — on paper, the strongest model tried this week by the metrics a non-clinical audience would look at first.
2. **Gradient Boosting trains and predicts far faster than Random Forest** (0.66 s vs 29.1 s to train; 0.006 ms vs 0.122 ms per prediction) while still improving macro recall over the baseline (0.405 vs 0.417 is roughly flat, but weighted F1 and accuracy both improve) — a cheaper compute profile than Random Forest for a similar interpretability cost.
3. **Both ensembles handle the everyday ESI 2–4 majority of patients better than the baseline**, which is where most ED throughput decisions actually happen day to day.

Three Arguments Against

1. **Random Forest misses every single ESI-1 patient in the test set — a recall of 0.000.** Its accuracy gain comes almost entirely from getting the common classes (ESI 2 and 3) right more often, which is exactly the aggregate-metric trap Tutorial 3 and Tutorial 4 warned about: the number that should decide this — not accuracy — got worse, not better.
2. **Tuning does not fix it**—A dedicated RandomizedSearchCV run scored explicitly on macro recall still produced 0.000 ESI-1 recall on the test set. This points to a structural limitation — majority-vote ensembles average away an extremely rare signal (77 of 55,121 encounters, 0.14%) — rather than a hyperparameter that was left in the wrong position. More tuning time would not likely change this conclusion.
3. **Gradient Boosting’s ESI-1 recall (0.375) is roughly half the baseline’s (0.750),** and it is the hardest of the three to explain per patient (SHAP was needed even to attempt an explanation — see decision journal). A model that is both less safe on the critical class and less transparent is a difficult sell to IT governance, regardless of its speed.

Risks and Unknowns

1. **Sixteen ESI-1 test patients is a very small sample**—A single patient’s outcome shifts ESI-1 recall by more than 6 percentage points, so none of these recall numbers should be treated as precise — only as a consistent, repeated pattern across three different tree-based configurations, which is what makes the “structural, not hyperparameter” conclusion credible.
2. **This dataset is still Yale-derived, not Caribbean**—Whatever a model’s recall on this data, it says nothing about performance at Mercer General until validated against local Emergency Department data — the Phase 1 argument this project has made since Week 5 stands regardless of which model wins this week’s benchmark.
3. **The SHAP explanation (decision journal) surfaced a case where the model’s output for a genuine ESI-1 patient looked clinically odd** — worth a joint review with Dr. Reyes before any model in this family goes further.

Recommendation

My recommendation is to:

1. Keep the Week 6 logistic regression baseline as the reference model for Phase 3
2. And not to promote Random Forest or Gradient Boosting on the strength of this week's results.

The right next step is not a fancier algorithm — it is fuller and better rare-class of data, which is precisely what Phase 1 of the Caribbean data foundation is meant to build before Phase 2 trains a locally validated classifier.

Complex models are worth revisiting once that foundation exists, and ESI-1 recall can be measured on more than sixteen patients.

Appendices

Supplementary Links

1. The GitHub link can be found [here](#)
2. The decision journal can be found in the docs for this week
3. Figures are included alongside this memo in the GitHub:
 - a. **figs/01–07_*.png**
 - b. **notebooks/Week7_Optimization_Model**

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	Meaning
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	Meaning
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol